ANDERSON SERANGOON JUNIOR COLLEGE

JC2 H2 Geography Preliminary Exam (2025)

H2 GEOGRAPHY **9173/ 01**

22 August 2025

3 hours

Additional Materials: Answer Booklet
1 Insert

READ THESE INSTRUCTIONS FIRST
If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your name and class on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer both questions in **Section A**.
Answer two questions in **Section B**, one from each cluster.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **both** questions.

Cluster 1: Development, Economy and Environment

1	Tiffany & Co. is a luxury jewellery transnational corporation (TNC) headquartered in the United States of America, with an extensive global production network (GPN). Resource 1 shows the distribution of global diamond supply per Million Carats. Resource 2 shows the distribution of primary sourcing and manufacturing of diamonds and metals of Tiffany & Co. in 2014. Resource 3 shows a factsheet on diamond mining in the Democratic Republic of Congo (DRC). Resource 4 shows an extract on sustainability efforts implemented by Tiffany & Co.		
	(a)	With reference to Resources 1 and 2, compare the spatial pattern of Tiffany & Co.'s primary source of rough diamond with the global diamond supply.	[4]
		• Both have Africa as the continent with the most number of countries with diamond supply/diamond sourcing (at least 6 countries for diamond supply, 3 countries for Tiffany's diamond sourcing). • Both have countries located on multiple continents across the globe. • However, global diamond supply is distributed across more continents and countries compared to Tiffany's diamond sourcing which is more limited. • However, Tiffany sources from countries that are relatively more developed, while diamond supply is distributed across both developed and less developed countries. • However, Tiffany only sources from countries of relatively higher latitudes.	
	(b)	Account for the global distribution of Tiffany & Co. manufacturing plants as seen in Resource 2.	[6]
		<u>Market Orientation</u> • Presence of manufacturing plants in different continents/regions to access, and cater to, different regional markets for sales of products • Presence of manufacturing plant in the United States → where Tiffany is headquartered → usually largest or most established consumer market + established brand image <u>Asset Orientation</u> • Presence of manufacturing plants in relatively less developed countries in Southeast Asia and Africa to tap on lower labour costs → lowering cost of factors of production to maximise profits • Located in close proximity to primary sources of diamonds (i.e. Namibia, Botswana) and raw materials (i.e. metals in USA) to tap on lower transportation costs to maximise profits	
	(c)	Explain how the GPNs of TNCs like Tiffany & Co. contribute to the interdependence between places in the global economy.	[4]
		<u>Trade flows</u> • Tiffany & Co. sources raw materials (e.g. diamonds from Namibia and Botswana) and sells finished jewellery in wealthy consumer markets like the USA, Europe, and Asia. This makes resource-exporting countries dependent	

		on demand from high-income markets, while consumer markets rely on steady global supply chains. <u>Capital flows</u> • Tiffany invests capital in manufacturing plants in Africa and Southeast Asia and retail outlets in Asia. Host countries depend on this FDI for jobs and infrastructure, while Tiffany relies on these locations to expand global reach and sustain profits. • Profits from overseas operations are repatriated back to the USA (or home country of TNCs), tying the financial performance of host economies to the corporate success of Tiffany's headquarters. This shows how capital circulates between places, making them financially interdependent. <u>Labour flows (and labour characteristics)</u> • Tiffany's GPN divides labour by skill: highly skilled designers and artisans in the USA, and semi-skilled workers in Botswana, Vietnam, or Mauritius for cutting, polishing, or assembly. Host countries depend on the jobs and wages provided, while Tiffany depends on their lower-cost yet skilled labour to remain competitive.	
	(d)	With the help of Resources 1, 2, and 3, describe the characteristics of extractive industries.	[4]
		• Extraction of non-renewable & locationally specific resources • Capital and technology intensive • Dominated by large private and/or state-owned firms *Support with evidence from Resources*	
	(e)	With reference to Resource 3 and 4, explain the impacts of the GPN of Tiffany & Co. on host economies.	[6]
		<u>Economic</u> • R4 shows that 'diamond industry employed over 15,000 people in 2023' in South Africa → showing how Tiffany's sourcing sustains jobs and incomes in host economies. This reduces unemployment and generates tax revenue for governments. • R4 also states that 'diamonds account for roughly 80% of exports and contribute 25% of GDP' in Botswana → meaning the national economy is heavily reliant on diamond sales to TNCs like Tiffany. This creates economic growth and foreign exchange earnings but also leaves Botswana vulnerable to fluctuations in global demand or Tiffany's sourcing decisions. <u>Social</u> • R3 states that 'mining exacerbates social insecurity (illicit trafficking, unsafe labour)' while the image shows harsh and dangerous working conditions with potential landslides or collapse of materials in the open pit mines → Host communities may experience instability if mining attracts black-market activity or undermines law enforcement, and while jobs are created, workers may be exposed to unsafe environments, reflecting uneven benefits of Tiffany's GPN. • R3 also shows that 'mining works as resource diversion from mitigating food insecurity → land, labour, and capital that could support agriculture and address food insecurity are instead channelled into diamond mining. For host	

		economies dependent on subsistence farming, this trade-off may worsen vulnerability to hunger and inequality. <u>Environmental</u> • R3 states that 'diamond mining water footprint in DRC is 15 to 35 times higher than irrigation water footprint', with R4 stating that 'mining displaces roughly 250 tons of earth per carat, polishing a single carat emits about 160kg of CO2, consumes 480L of water, and uses 52 kWh of energy'. → highlights how GPN activities can place extreme pressure on scarce local water resources the significant environmental degradation and carbon intensity imposed on host environments • However, R4 also shows that 'Tiffany has cut its direct and indirect carbon emissions by 33% since 2019' and 'around 12% of its global floor area in buildings' are sustainably designed → reflects a commitment to greener operations. When Tiffany applies similar standards to facilities in host economies, it can transfer green technology and best practices, helping reduce resource use and environmental degradation locally.	
	(f)	Suggest how states can influence the operations of TNCs like Tiffany & Co. through their roles as regulators of economic activities.	[6]
		• Market Regulation (ensuring fairness and competition) ○ States enforce competition laws to prevent monopolies and protect consumers. ○ Influence on TNCs: TNCs must adapt pricing, mergers, and partnerships to avoid penalties. ○ Example: The Competition and Consumer Commission of Singapore (CCCS) fined Grab in 2018 after its merger with Uber reduced competition in the ride-hailing market. This shows how state regulation directly shapes TNC market behaviour. • Regulation of Economic Flows (capital and labour) ○ States control flows of capital (e.g., profit repatriation rules, capital controls) and labour (e.g., work visas). ○ Influence on TNCs: Restrictions can delay investments, reduce liquidity, or limit the mobility of skilled workers across subsidiaries. ○ Example: In 2015, Greece imposed capital controls during its debt crisis, restricting how firms (including TNCs) could move money out of the country, disrupting financial operations. • Regulation of Trade Policies ○ Governments use tariffs, quotas, and product standards to manage imports and exports. ○ Influence on TNCs: Tariffs raise costs, disrupt supply chains, and may force relocation of production closer to markets. ○ Example: The US–China trade war led to higher tariffs on goods like electronics and luxury products, forcing TNCs to diversify supply chains or pass higher costs to consumers. • Foreign Direct Investment (FDI) Policies ○ States regulate entry rules for foreign firms, sometimes requiring joint ventures or setting ownership caps. ○ Influence on TNCs: TNCs may have to share technology with local firms or adjust expansion strategies.	

		○ Example: In China, restrictions on wholly foreign-owned enterprises historically required joint ventures, shaping how TNCs in industries like automotive or retail entered the market. • Industry Policies (financial incentives, employment, and legislation) ○ States provide incentives (e.g., subsidies, tax breaks) or impose obligations (e.g., minimum wages, sustainability laws). ○ Influence on TNCs: This affects location decisions, production costs, and compliance strategies. ○ Example: China's creation of Special Economic Zones (e.g., Shenzhen SEZ) with tax incentives and relaxed regulations attracted TNCs to set up manufacturing and R&D hubs, directly shaping global production networks.	

Cluster 2: Tropical Environments

2	Resource 5 shows the movement of Tropical Storm Washi across the Philippines from 15 December to 19 December 2011. Resource 6 shows fluvial flooding at Cagayan de Oro City and its location in the Cagayan River Basin. Resource 7 shows a flood prevention measure along a stretch of the Cagayan River.		
	(a)	Describe the path and changes in storm intensity of Tropical Storm Washi as it moved across the Philippines, as shown in Resource 5.	[4]
		Direction of movement • It moved generally from **east to west**, originating in the western Pacific, passing over the Philippines, and continuing into the South China Sea. • While moving broadly in a westward path, the storm occasionally **shifts its course**, veering northwest on Dec 16th to 18th and southwest on Dec 18th. **Changes in intensity** • It began as a tropical depression until 15 December, then intensified into a tropical storm, maintaining that strength until 19 December, before weakening back into a tropical depression.	
	(b)	With reference to Resource 5, suggest possible reasons for the formation of Tropical Storm Washi.	[5]
		• Intense insolation warms sea surface temperatures in the Pacific Ocean, often exceeding 26.5°C, providing the heat and moisture necessary for storm formation. This heating drives evaporation, creating water vapor and a low-pressure cell that draws winds from higher-pressure areas. Rapidly rising warm air forms cumulonimbus clouds, and the release of latent heat during condensation warms the surrounding air, increasing its buoyancy. This enhanced buoyancy causes the air to rise further, drawing in additional moist air from below and fueling the storm's growth. • Large expanse of surrounding warm ocean – With vast open waters of the Western pacific , storms have ample space to develop and intensify before making landfall. Moreover, the Western Pacific near the Philippines often experiences warmer sea surface temperatures during La Niña. These variations in ocean currents and temperatures strongly influence storm formation and strength in the region. • Cyclones require low vertical wind shear (minimal change in wind speed and direction with height); otherwise, the storm structure is disrupted and torn apart. In the tropics, vertical wind shear is generally weaker because large-scale circulation (e.g., the Hadley Cell) is dominated by rising and sinking air motions, rather than by the strong horizontal temperature contrasts that drive the powerful jet streams of the mid-latitudes. • The latitudinal range of about 5°–15°N is favourable for cyclone formation, as the Coriolis effect is strong enough in this zone to induce the rotational spin of clouds, leading to the development of cyclonic systems.	
	(c)	Using Resources 5 and 6, explain the factors that led to fluvial flooding in Cagayan de Oro City on December 17, 2011.	[6]
		• The passage of Tropical Storm Washi brought torrential rainfall contributing large volumes of direct channel precipitation to the Cagayan River. At the same time, high rainfall intensity (≥350mm) exceeded the city's infiltration capacity, producing Hortonian overland flow that drained into the river. This combined input raised the river discharge beyond its bankfull capacity, causing it to overflow and flood the city.	

		• The fan-shaped basin, characterised by high drainage density and numerous tributaries, funnels stormwater rapidly into the main river. This causes discharge to rise quickly, increasing the risk of flooding—particularly near the mouth of the Cagayan River, where multiple flows converge and water levels rise sharply. • The meandering course of River Cagayan, with its sharp bends, reduces the flow velocity as water must travel a longer, winding path. This slows the downstream movement of storm waters, preventing rapid discharge and increasing the likelihood of water backing up and overflowing onto the floodplain. • Th built-up areas of the city create extensive impermeable surfaces (e.g. roofs, car parks, roads and pavements). These prevent infiltration, leading to rapid generation of overland flow. The resulting stormwater quickly drains into the river, raising discharge levels and increasing the risk of flooding. • Land clearance removes protective vegetation cover, removing the crucial role of canopy interception. Consequently, rainfall strikes the exposed ground surface unimpeded, with high-energy raindrops causing rainsplash erosion. This process dislodges soil particles, which subsequently clog surface pores (surface sealing). The sealed surface significantly reduces soil infiltration capacity, thereby promoting rapid overland flow. This increased volume and velocity of overland flow converges into river channels, causing a sharp rise in discharge that frequently exceeds bank capacity and leads to flooding.	
	(d)	Describe the circumstances under which deposition may take place within a river.	[5]
		Physical settings favouring deposition within a river • Deposition happens on the **inside bend of a meander** where water moves slowly. Centrifugal force pushes the fastest current to the outside, causing erosion there. The slower water on the inside loses the energy to carry sediment, which settles as sand and gravel, forming a slip-off slope or point bar. • When a **river enters a standing body of water,** its velocity drops sharply. Because sediment-carrying capacity depends on water speed, the river first deposits the heaviest and coarsest particles, like sand and gravel, closest to the river mouth. Finer particles, such as silt and clay, are carried farther before settling. Over time, this sorting of sediments builds up a delta, with coarser materials near the river channels and finer layers forming further out into the lake or sea. • Deposition occurs where the **channel gradient decreases** because a gentler slope reduces the river's velocity and energy. In steep mountain channels, rivers flow quickly and can carry large amounts of sediment. When the river reaches a flat plain, it slows down, losing the energy needed to transport its sediment load. As a result, sediments—starting with the heaviest, like boulders and gravel—are deposited first, followed by sand, silt, and clay further downstream. This process often forms alluvial fans at the base of mountain ranges. • Deposition occurs when **water volume decreases** because the river's capacity to carry sediment depends on the amount of water and its energy. When the water volume drops—due to drought, seasonal changes, or diversion—the flow slows and loses the energy needed to transport sediment. As a result, sediments such as sand, silt, and gravel settle out, accumulating on the riverbed or along the banks. • Deposition occurs when a river has a **high sediment load** because it may exceed the river's capacity to carry all the material. When the flow cannot transport all the sediment, the excess settles out, forming deposits along the	

		riverbed, banks, or at the river mouth. This often happens after heavy rainfall, rapid erosion upstream, or in rivers flowing through easily erodible terrain. • When a river encounters obstacles such as boulders, vegetation, or artificial structures, it slows down and loses energy, causing sediment to settle. Human-made structures like dams create reservoirs where water velocity drops almost to zero, resulting in extensive sediment deposition upstream.	
	(e)	Explain the benefits and potential limitations of the flood prevention measure along the Cagayan River shown in Resource 7.	[5]
		<u>Benefits</u> • A floodwall acts as a sturdy barrier that physically block floodwaters from reaching communities, infrastructure, and agricultural land, greatly reducing the risk of property damage and loss of life. • Effective in urban areas with densely built environments, as they require less space than levees and can protect valuable real estate, such as city centers and industrial zones. • Their rapid deployment and durability also make them a practical solution for both temporary and long-term flood protection, ensuring cities and towns remain functional and safe during high river flows. • Often more economical than relocating entire communities or infrastructure <u>Potential limitations</u> • Risk of overtopping or failure: If a flood exceeds the design level, floodwalls can be overtopped or breached, potentially causing severe damage—the same as, or worse than, if no protection was present. May collapse if foundations are undermined by erosion or seepage • High costs and maintenance: Building and maintaining floodwalls, especially tall ones, can be expensive. They require regular inspections and upkeep to remain effective. • Limited protection scope: Floodwalls are usually suited for certain flood types or sizes and may provide insufficient protection against large or prolonged flood events. • Drainage Issues: Can trap local rainwater behind the wall, causing interior flooding • False sense of security: Floodwalls can create a false sense of safety, leading communities to underestimate residual risks, particularly as climate patterns change and extreme events become more common. Can create overconfidence leading to inappropriate development • Flood transfer: Redirects floodwater to unprotected areas downstream or across the river	
	(f)	Explain the formation of river cliffs and point bars.	[5]
		<u>River cliffs</u>: • Formed on the outside banks of meanders, where the channel curves, causing the faster-moving part of the river—the thalweg—to swing toward the bank. • At these outer bends, water velocity and energy are greatest, creating strong erosive forces that actively wear away the bank material. • Main processes responsible are hydraulic action (the force of water hitting the bank), abrasion (sediment scraping and scouring the bank) and cavitation (pressure changes causing bubbles that break apart soil and rock).	

		• Continuous erosion weakens and undercuts the bank at the meander's outer edge, sometimes causing collapses or slumping, which further steepens the bank and leads to the distinctive vertical or overhanging river cliff. **Point bars**: • Formed on the inside bend of a meander, where conditions encourage deposition instead of erosion. • On the inner bend, the water travels a shorter distance, so it moves more slowly. In addition, the flow on the inside is shallower and comes into contact with the riverbed and bank more, which creates greater friction. This friction reduces the water's velocity and energy, leading to slower flow. • With lower velocity, the river loses transport capacity, so it drops its load. Mostly sand, silt, and fine sediment settle out here. • The deposited material builds up into a gently sloping feature known as a point bar or slip-off slope. This slope extends gradually from the channel up towards the floodplain.	

Section B

Answer **either** question 3 **or** question 4, and answer **either** question 5 **or** question 6.

Cluster 1 Development, Economy and Environment

3	Evaluate the validity of Malthus' and Boserup's views in explaining the relationship between population and provisioning services today.	[20]
	Candidates could adopt the approach that both Malthus' and Boserup's theories retain partial relevance when evaluating the population-resource relationship. Malthus' pessimistic theory, which argues that population growth may outpace the arithmetic rise of food supply, is supported by examples of resource scarcity, famine, and poverty in areas where environmental limits and conflict prevail. This aligns with Neo-Malthusian concerns over finite resources and the risk of overshoot. Conversely, candidates may highlight Boserup's optimistic perspective, which suggests that population pressure drives technological innovation and agricultural intensification—a concept validated by case studies such as the Green Revolution and ongoing advances in global food production. Ultimately, candidates should recognize that contemporary reality is complex: Malthusian limitations—such as finite resources and inequalities in distribution—can coexist with Boserupian adaptations, where human ingenuity enables solutions to scarcity. A nuanced argument will show that local or regional Malthusian crises may arise even as broader, Boserupian advancements support continued population growth on a global scale	
4	Evaluate the view that transnational corporations (TNCs) are considered a mixed blessing to the host economies where they locate their production.	[20]
	Candidates could adopt a balanced structure that defines the "mixed blessing" by dedicating equal analysis to both the benefits and the costs of Transnational Corporations (TNCs) on host economies. The argument should first highlight the essential advantages, such as providing significant Foreign Direct Investment (FDI), creating jobs, facilitating technology transfer, and improving infrastructure. This should be immediately juxtaposed with the serious drawbacks, including the repatriation of profits (leading to capital flight), the potential exploitation of local labor through low wages, environmental degradation, and the crowding out of smaller domestic firms. The evaluation must conclude by arguing that the net impact is highly conditional, dependent on the regulatory strength and policy framework of the host government to ensure the positive spillovers outweigh the risks of dependency and corporate political influence.	

Cluster 2 Tropical Environments

5	'The impacts of fluvial floods in the humid tropics are always negative.' Evaluate this statement.	[20]
	Possible Approaches: Candidates could approach the question by making a judgement on whether the impacts of fluvial floods in the humid tropics are always negative through a consideration of the range of impacts of fluvial floods in the humid tropics. Candidates could evaluate whether the impacts are exclusively detrimental, or if some could be beneficial to the environment,	

	economy, or society in certain contexts. Candidates could also evaluate which impacts are most significant in shaping the overall perception of floods as negative or otherwise. Candidates could also approach the question by making a judgment on whether the impacts of fluvial floods in the humid tropics are always negative through a consideration of two or more case studies. Candidates could evaluate whether differences in preparedness, land use, and socio-economic conditions influence the balance between positive and negative outcomes. Candidates could also consider contextual factors such as flood magnitude, frequency, and the adaptive capacity of communities, which may contribute to differences in whether the impacts can be seen as wholly negative or partly beneficial. *Levels marked using Generic Level Descriptors for H2 essays.*	
6	'The movement of materials on slopes is largely influenced by natural factors.' Evaluate this statement.	[20]
	Possible Approaches: Candidates could approach the question by making a judgement on whether the movement of materials on slopes is influenced by natural factors through a consideration of the relative significance of natural factors vis-à-vis human factors. Candidates could analyse natural factors such as prolonged rainfall and earthquakes, which could trigger a sudden movement of materials. Candidates could also analyse both natural and human factors which may contribute to movement of materials on slopes over time such as weathering and deforestation. Candidates could even analyse human factors such as slope management strategies which could reduce the occurrence of movement of materials on slopes. Candidates could also approach the question by making a judgment on whether some factors have a greater influence in some contexts than others through a consideration of two or more case studies. Candidates could evaluate the relative significance of the different factors in each case study and analyse the interplay between the factors. *Levels marked using Generic Level Descriptors for H2 essays*	